Lecture plan · for the lecturer

Planning and tool use with language models

AI agents · University · Lecture 2

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Plan with A*.
- Build ReAct agents.
- Ground answers in tools.

Activity: Search, then act

1. Warm-up (5 min): Show the invented-room transcript and ask what step is missing.
2. Trace (20 min): Pairs trace A* on a six-node map with the open list on paper.
3. Debug (15 min): Find the missing tool call in a ReAct transcript and rewrite the step.
4. Compare (15 min): List what A* guarantees that the agent does not.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- When is A* optimal?
- Why did the agent skip the tool?
- How will you grade transcripts?

Exit ticket

- Write one instruction that grounds an agent's answers.

Try it in your next lecture
Trace A* on a six-node campus map on the board, then show a ReAct transcript with one missing tool call and ask pairs to find the step where the agent should have acted.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **Which change most directly stops the agent inventing a room?**

◯ A bigger model ◯ Require and check a tool result ◯ A longer answer

2 **Expand A*.**

Node X has g = 4 and h = 3; node Y has g = 2 and h = 6. A* expands _____ first because its f is _____ .

Word bank: Y · X · 7 · 8

3 **Which reduce hallucinated answers in a tool-using agent? Choose all that apply.**

☐ Clear, specific tool descriptions ☐ Validating answers against tool output

☐ Instructions to answer only from tool results ☐ Removing the step limit

4 **Unlike A* with an admissible heuristic, a ReAct agent gives no guarantee that its plan is optimal.**

◯ True ◯ False

5 **Order one ReAct step.**

☐ Action: call a tool

☐ Answer or next thought

☐ Thought: what is needed

☐ Observation: read the result

Answers · for the lecturer

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1 **Require and check a tool result**

Ground the answer in what the tool returned.

2 **Node X has g = 4 and h = 3; node Y has g = 2 and h = 6. A* expands X first because its f is 7.**

f(X) = 7 < f(Y) = 8.

3 **Clear, specific tool descriptions; Instructions to answer only from tool results; Validating answers against tool output**

Removing step limits invites loops; it does not ground answers.

4 **True**

Language-model agents offer no optimality or completeness guarantees.

5 **Top to bottom: 2, 4, 1, 3**

Reasoning is grounded by reading the observation before continuing.

For you to think about

Do your students learn classical search before they build language-model agents? What would they miss without it?